

Summary

Senior Machine Learning Researcher focused on trustworthy AI, secure LLM agents, and AI-for-security systems.

Lead a team of four Ph.D. researchers on controllable security data synthesis; also research AI-agent permission management and federated-learning security.

Research spans LLM safety, agent governance, efficient inference, distributed AI systems, and networked security.

Technical Skills

AI/ML	Python, PyTorch, Transformers, LLM agents, SLM adaptation, diffusion models, federated learning
Systems	KV-cache optimization, model serving, distributed systems, networking, low-latency inference
Security	AI firewall, adversarial robustness, model poisoning, backdoors, content safety, policy enforcement

Work Experience

Edison Research Center, Huawei Canada

Jan. 2026 – Present

Senior Machine Learning Researcher

Canada

- Lead a team of four Ph.D. researchers developing controllable data synthesis methods for AI security applications
- Design a GAN-inspired framework that pairs a diffusion language model generator with an encoder-only discriminator to steer generated data toward target styles
- Develop permission-management mechanisms for AI agents to govern tool access and resource usage across agentic workflows
- Investigate model-poisoning attacks and defense strategies in federated learning to improve robustness against malicious clients

Edison Research Center, Huawei Canada

Dec. 2024 – Dec. 2025

Machine Learning Researcher

Canada

- Developed an AI web application firewall using compact BERT-family models to detect malicious and policy-violating inputs at the application boundary
- Trained domain-specific tokenizers and small language models for accurate, low-latency security classification
- Designed a multimodal phishing-email detection agent that extracts evidence from message content, attachments, and embedded URLs, then orchestrates LLM calls for efficient threat assessment

6G Network Architecture Design

Dec. 2020 – Jul. 2023

Ph.D student

Victoria, Canada

- Self-Evolving and Transformative Protocol Architecture for 6G design
- Updatable eXpress Transport Protocol design
- Delay-Guaranteed Routing Protocol design

Computing Research Institute of Aviation Industry Corporation of China

Dec. 2018 – Dec. 2019

Researcher

Xi'an, China

- Sub-6G (4 200 MHz - 4 400 MHz)channel modeling
- Wireless communication system design based on Universal Software Radio Peripheral (USRP)
- Excellence Research Intern Award Recipient

Toward the 6G network architecture
Wireless Networking Technology Forum Huawei Canada, Victoria, Canada

July 2022

- Presented key design principles and challenges for the development of 6G networks, including ultra-low latency, massive connectivity, and enhanced security. Discussed the role of AI-driven network management and potential use cases such as smart cities and autonomous systems.

Multi-path routing protocol to support Quality of Service
IEEE MASS 2023, Toronto, Canada

Sep, 2023

- Discussed methods to improve network stability in case of congestion and network jamming through the application of multi-path routing protocols, emphasizing their ability to maintain service quality under varying traffic conditions.

Deadline Driven Routing Protocol
IEEE Conference on Computer Communications, Vancouver, Canada

May 2024

- Presented the Deadline Driven Routing (DDR) protocol, designed to guarantee service with strict delay constraints in real-time communication systems. Discussed the design, implementation, and evaluation of DDR, focusing on its performance in achieving delay guarantees and improving reliability in delay-sensitive applications.

Enhancing Cloud Computing with Networking Technique
GenAI and Future Network Forum (Ericsson), Victoria, Canada

July 2024

- Explored the integration of advanced networking techniques to enhance the efficiency and scalability of cloud computing infrastructures. Demonstrated how novel routing protocols and optimization strategies can improve cloud service delivery, particularly in large-scale distributed environments.

Professional Service

Reviewer

IEEE INFOCOM, IEEE/ACM Transactions on Networking (ToN), Computer Networks, IEEE Internet of Things Journal, and ACM SIGKDD (KDD)

Awards & Scholarships

2024

Chinese Government Award for Outstanding Self-financed Students Abroad

2020–2023

UVic Graduate Fellowship; Excellent Research Intern Achievement

2017–2019

NWPU Graduate Scholarship; ACM-ICPC Second Prize of NWPU (2013)

Selected Publications

- **P. Yang**, T. Chang, L. Cai, “DDR: A Deadline-Driven Routing Protocol for Delay Guaranteed Service” IEEE International Conference on Computer Communications (INFOCOM), Jan 15. 2024.
- **P. Yang**, L. Cai, T. Chang, “DGR: Delay-Guaranteed Routing Protocol” The 20th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), July. 2023.
- X. Ren, L. Cai, **P. Yang**, and J. Ji, “Congestion-aware Delay-guaranteed Scheduling and Routing with Renewal Optimization,” Computer Networks, June 2023.
- X. Fan , S. Li , W. Cui , **P. Yang**, G. Huang, ”Mixed-Numerology Channel Division for Wireless Avionics Intracommunications”, Wireless Communications and Mobile Computing, April. 2022